



DEPARTMENT OF COMPUTER SCIENCE
FACULTY OF PHYSICAL SCIENCES
UNIVERSITY OF BENIN, BENIN CITY

2024/2025 BSC COMPUTER SCIENCE FIRST SEMESTER EXAMINATIONS
COURSE: IFT211 - DIGITAL LOGIC AND DESIGN

Time: 60 Minutes

INSTRUCTIONS: Attempt all Questions and shade the correct option corresponding to the answer chosen.

1. Which gate is known as the universal gate? (a) NAND (b) OR (c) AND (d) XOR (e) None of the above
2. How does a logic gate function? (a) Using the flow of electric current (b) Spontaneously (c) By using energy from a chemical reaction (d) All of the above (e) None of the above
3. Which of the following is not a logic gate? (a) AND (b) OR (c) IF (d) NOT (e) All of the above
4. Which of the following signifies a NOT gate? (a) 1 (input)- 0 (output) (b) 1,0 (input)- 1 (output) (c) 0,0 (input)- 0 (output) (d) 1,1(input)-0 (output) (e) None of the above
5. Which of the following gates can function on a single input? (a) NOT (b) AND (c) OR (d) NOR (e) None of the above
6. A NAND gate has — inputs and — outputs. (a) High input and High output (b) High input and low output (c) Low input and low output (d) Low input and high output (e) None of the above
7. A Truth Table is a ... (a) table representing the output one will get for a given input (b) table representing what input and output will be given by a given boolean operator (c) mathematics table used to determine the output of logic (d) fundamental tool in digital electronics, computer science, and logic (e) All of the above
8. How can a full adder be implemented using half adders? (a) 1 half adder (b) 2 half adders and 1 OR gate (c) 3 half adders and 2 OR gates (d) Only with full adders (e) None of the above
9. What combination is a NAND gate? (a) NOT AND (b) NOT OR (c) NOT NOT (d) XOR NOR (e) None of the above
10. A minterm is defined as (a) A product of all variables in complemented form (b) A product of all variables in either complemented or uncomplemented form (c) sum of all variables in complemented form (d) sum of all variables in either complemented or uncomplemented form (e) None of the above
11. A maxterm is defined as the (a) sum of all variables in either complemented or uncomplemented form (b) product of all variables in complemented form (c) product of all variables in either complemented or uncomplemented form (d) sum of all variables in uncomplemented form only (e) All of the above
12. The minterm for the combination $A = 1, B = 0, C = 1$ is: (a) $A + B' + C$ (b) $A'B'C'$ (c) $AB'C$ (d) ABC' (e) None of the above
13. The maxterm for the combination $A = 1, B = 0$ is: (a) $A + B$ (b) $A + B'$ (c) $A' + B$ (d) $A' + B'$ (e) None of the above
14. Which of the following represents a canonical sum of products (SOP) form? (a) $AB + A'B'$ (b) $AB + AC$ (c) $A + B + C$ (d) $A'B'C + AB'C + ABC$ (e) None of the above
15. Which of the following is a sequential circuit? (a) Multiplexer (b) Decoder (c) Flip-Flop (d) Adder (e) None of the above
16. What distinguishes a sequential circuit from a combinational circuit? (a) Use of only logic gates (b) Dependence on current inputs only (c) Dependence on both current inputs and previous states (d) It has no outputs (e) None of the above
17. Which memory element is commonly used in sequential circuits? (a) Resistor (b) Capacitor (c) Flip-Flop (d) Inductor (e) None of the above
18. What type of sequential circuit does not use a clock signal? (a) Synchronous (b) Asynchronous (c) Combinational (d) Mixed (e) None of the above

19. The output of a flip-flop is dependent on: (a) Only the input at that time (b) The input and the previous output (c) The clock frequency only (d) Only the previous output (e) None of the above
20. What is the main drawback of an SR flip-flop? (a) It cannot store data (b) It has a forbidden input condition (c) It operates without a clock (d) It consumes too much power (e) None of the above
21. Which of the following flip-flops is free from the invalid state? (a) SR flip-flop (b) JK flip-flop (c) D flip-flop (d) T flip-flop (e) None of the above
22. In a D flip-flop, what happens when the clock triggers? (a) Q becomes 0 (b) Q becomes 1 (c) Q copies the D input (d) Q toggles (e) None of the above
23. What is the function of the clock signal in a sequential circuit? (a) To process inputs (b) To maintain voltage (c) To synchronize changes in the circuit (d) To generate outputs (e) None of the above
24. What is a combinational circuit? (a) A circuit with feedback loops (b) A circuit whose output depends only on the present inputs (c) A circuit that stores previous states (d) A sequential circuit (e) None of the above
25. A full adder has how many inputs and outputs? (a) 3 inputs, 2 outputs (b) 2 inputs, 1 output (c) 3 inputs, 1 output (d) 2 inputs, 2 outputs (e) None of the above
26. Which gate is used in the design of a half adder? (a) AND and OR gates (b) OR and XOR gates (c) XOR and AND gates (d) NAND and XOR gates (e) None of the above
27. Which of the following is NOT a combinational circuit? (a) Decoder (b) Encoder (c) Flip-Flop (d) Multiplexer (e) None of the above
28. Which of the following is the identity law in Boolean algebra? (a) $A + 0 = A$ (b) $A + A = A$ (c) $A + 1 = 1$ (d) $A \cdot A = A$ (e) None of the above
29. What is the value of $A + A'$? (a) A (b) 0 (c) 1 (d) A' (e) None of the above
30. Which Boolean expression represents the AND operation? (a) $A + B$ (b) $A \cdot B$ (c) $A \oplus B$ (d) A' (e) None of the above
31. What is $A \cdot 0$ equal to? (a) 0 (b) 1 (c) A (d) A' (e) None of the above
32. What is the dual of the Boolean expression $A + 0 = A$? (a) $A \cdot 0 = A$ (b) $A \cdot 1 = A$ (c) $A + 1 = A$ (d) $A + A = A$ (e) None of the above
33. What is the binary equivalent of the decimal number 13? (a) 1100 (b) 1011 (c) 1101 (d) 1110 (e) None of the above
34. What is the Boolean expression for a NAND gate? (a) $A + B$ (b) $A \cdot B$ (c) $(A \cdot B)'$ (d) $A \oplus B$ (e) None of the above
35. Which of the following is a basic logic gate? (a) AND (b) OR (c) NOT (d) All of the above (e) None of the above
36. What is the result of the binary addition $1011 + 1101$? (a) 11000 (b) 10100 (c) 10000 (d) 11100 (e) None of the above
37. Which gate gives a high output of (1) when any one of its inputs is high? (a) AND (b) OR (c) NOT (d) NAND (e) None of the above
38. Which gate is a NOT universal gate? (a) RAND (b) ORAND (c) NORAND (d) NOTAND (e) All of the above
39. How many inputs and outputs does a half adder have? (a) 1 input, 2 outputs (b) 2 inputs, 1 output (c) 2 inputs, 2 outputs (d) 3 inputs, 2 outputs (e) None of the above
40. Which of the following limitations does a half adder have? (a) Cannot add binary numbers (b) Cannot add two bits (c) Cannot handle carry from a previous stage (d) All of the above (e) None of the above



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2024/2025 BSC COMPUTER SCIENCE SECOND SEMESTER EXAMINATIONS

COURSE: CYB201 - CYBERSECURITY AND STRATEGY

CREDIT UNIT: 02

TIME: 50 MINUTES.

INSTRUCTION: Answer all the Questions by shading the corresponding correct answers on YOUR OMR sheet.

1. What does the CIA Triad stand for in information security? A) Confidentiality, Integrity, Accessibility B) Control, Integrity, Availability C) Confidentiality, Integrity, Availability D) Compliance, Integrity, Authentication E) Confidentiality, Identification, Availability
2. Which encryption standard uses the same key for encryption and decryption? A) RSA B) AES C) SHA-256 D) TLS E) OAuth
3. A hospital uses AES encryption for patient records and maintains offline backups. Which principle of the CIA Triad does this primarily address during a ransomware attack? A) Confidentiality B) Integrity C) Availability D) Authentication E) Non-repudiation
4. Data classified as "Secret/Restricted" requires which security measure? A) Basic access controls B) Public sharing on the company website C) Multi-factor authentication and isolated networks D) TLS for data in transit only E) Weekly vulnerability scans
5. Which method ensures data on a decommissioned hard drive is irretrievable? A) Formatting the drive B) Cryptographic erasure C) Using a firewall D) Installing antivirus software E) Enabling MFA
6. Which protocol is used for enterprise Single Sign-On (SSO)? A) HTTP B) SAML C) WPA3 D) AES E) DNSSEC
7. A penetration tester with no prior knowledge of the target system is performing: A) White Box Testing B) Grey Box Testing C) Black Box Testing D) SAST E) DAST
8. Phishing simulation exercises aim to: A) Test network bandwidth B) Train employees to recognize social engineering attacks C) Encrypt sensitive emails D) Update antivirus signatures E) Generate threat intelligence reports
9. A VLAN is used for: A) Network segmentation B) Encrypting emails C) Multi-factor authentication D) Generating hashes E) Backing up data
10. A vendor security assessment is part of: A) Third-Party Risk Management (TPRM) B) Incident Response C) Cryptographic erasure D) Data classification E) TLS implementation
11. Security awareness training reduces risks associated with: A) Hardware failures B) Social engineering C) AES key management D) Firewall misconfigurations E) Data redundancy
12. A Web Application Firewall (WAF) protects against: A) SQL injection B) Data theft from decommissioned drives C) Insider threats D) Weak encryption keys E) Phishing emails
13. STRIDE is a methodology for: A) Threat modeling B) Calculating ALE C) Securing APIs D) Network segmentation E) Data classification
14. JWT (JSON Web Tokens) are often used for: A) API authentication B) Encrypting databases C) Securing DNS D) Data erasure E) Security awareness
15. Which control ensures users have minimal necessary access? A) Least Privilege B) Defense in Depth C) Redundancy D) Hashing E) Salting
16. A file's SHA-256 hash changes if: A) Its name is modified B) Its content is altered C) It is moved to a new folder D) It is encrypted with AES E) It is uploaded to the cloud
17. Which protocol is used for secure remote server access? A) SSH B) HTTP C) DNS D) FTP E) SNMP
18. An IPS (Intrusion Prevention System) can: A) Block malicious traffic in real-time B) Encrypt data at rest C) Generate threat intelligence reports D) Classify data E) Train employees
19. "Data at rest" refers to information: A) Transmitted over a network B) Stored on a hard drive C) Processed in memory D) Displayed on a screen E) Deleted via cryptographic erasure
20. A phishing simulation targets: A) Network bandwidth B) Employee vigilance C) Firewall rules D) Data classification E) Backup integrity
21. Which of the following is NOT one of the three categories of cybersecurity methods mentioned in the notes? A) Preventive measures B) Detective measures C) Corrective measures D) Predictive measures E) All of the above are cybersecurity methods
22. What is the main difference between a virus and a worm? A) Viruses are more dangerous than worms B) Worms are larger in file size than viruses. C) Viruses require a host program to execute, worms do not D) Worms only affect Windows systems, viruses affect all systems E) There is no difference between viruses and worms
23. Which fault-tolerant methodology involves using different types of components for protection? A) Redundancy

- B) Diversity C) Isolation D) Authentication E) Non-repudiation
24. What does "non-repudiation" mean in cybersecurity? A) Preventing unauthorized access to information B) Ensuring authorized users have access when needed C) Verifying the identity of users or devices D) Ensuring users cannot deny having performed an action E) Protecting information from unauthorized modification
25. Which of the following is an example of the "availability" cybersecurity element? A) A company encrypts its sensitive data B) A bank uses digital signatures for transactions C) A website uses a CDN to handle large numbers of visitors D) A user enters username and password to log in E) A company uses a firewall to block unauthorized access
26. The WannaCry ransomware attack infected approximately how many computers and in how many countries? A) Over 100,000 computers in 100 countries B) Over 200,000 computers in 150 countries C) Over 300,000 computers in 200 countries D) Over 50,000 computers in 75 countries E) Over 500,000 computers in 250 countries
27. What is the difference between authentication and authorization? A) Authentication verifies identity, authorization grants or denies access to resources B) Authorization verifies identity, authentication grants or denies access to resources C) They are the same thing with different names D) Authentication is for users, authorization is for devices E) Authentication is preventive, authorization is detective
28. Which of the following is NOT a direct advantage of cybersecurity in an organization A) Protects sensitive information B) Maintains the integrity of data C) Ensures the availability of systems and data D) Builds trust with customers and partners E) Reduces operational costs for organizations
29. Which type of security policy defines appropriate use of company resources, including computers, internet, and email? A) Access Control Policy B) Acceptable Use Policy (AUP) C) Information Security Policy D) Incident Response Policy E) Data Protection & Encryption Policy
30. What are the six phases of the NIST Incident Response Lifecycle in the correct order? A) Detection, Preparation, Containment, Eradication, Recovery, Lessons Learned B) Preparation, Detection & Analysis, Containment, Eradication, Recovery, Lessons Learned C) Containment, Detection, Preparation, Eradication, Recovery, Analysis D) Preparation, Containment, Detection, Recovery, Eradication, Lessons Learned E) Detection, Containment, Preparation, Recovery, Eradication, Lessons Learned
31. In the Target Data Breach (2013), what was the primary security failure that led to the compromise? A) Employees clicked on phishing emails B) The company failed to encrypt customer data C) Target failed to enforce strict third-party access controls D) The company used weak passwords E) Target's firewall was misconfigured
32. Which cybersecurity best practice principle states that employees should have the minimum access necessary for their jobs? A) Multi-Factor Authentication (MFA) B) Defense in Depth C) Principle of Least Privilege (PoLP) D) Zero Trust Architecture E) Security by Design
33. What type of security testing involves ethical hackers simulating real-world cyberattacks to test system defenses? A) Vulnerability Scanning B) Security Audits C) Code Review D) Penetration Testing (Ethical Hacking) E) Social Engineering Testing
34. Which security testing method uses automated tools to detect system vulnerabilities? A) Penetration Testing B) Vulnerability Scanning C) Security Audits D) Social Engineering Testing E) Code Review
35. software vulnerabilities C) Improper enforcement of multi-factor authentication (MFA) D) Lack of employee training E) Misconfigured firewall rules
36. Which of the following may NOT be listed as a key cybersecurity best practice A) Multi-Factor Authentication (MFA) B) Regular Security Updates & Patch Management C) Employee Cybersecurity Training D) Blockchain Implementation E) Principle of Least Privilege (PoLP)
37. According to the NIST Incident Response Lifecycle, what is the correct order of phases? A) Detection, Preparation, Containment, Recovery, Eradication, Lessons Learned B) Preparation, Detection & Analysis, Containment, Eradication, Recovery, Lessons Learned C) Containment, Detection, Preparation, Eradication, Recovery, Analysis D) Preparation, Containment, Detection & Analysis, Recovery, Eradication, Lessons Learned E) Detection & Analysis, Preparation, Containment, Recovery, Eradication, Lessons Learned
38. In the Colonial Pipeline Ransomware Attack (2021), what was the primary security weakness that allowed the attackers to gain access? A) Unpatched software vulnerabilities B) A compromised single employee password C) Weak firewall configuration D) Lack of network segmentation E) Inadequate backup systems
39. What are the three main goals of cybersecurity? A) Prevention, Detection, Response B) Confidentiality, Integrity, Availability C) Authentication, Authorization, Accounting D) Encryption, Firewalls, Antivirus E) none of the above
40. The three main types of access control are A) Mandatory access control, Discretionary main control and Role-play access control B) Discretionary main control, Mandatory based control, and Role-based access control C) Mandatory access control, Integrity based Control and Discretionary based control D) Mandatory Access based control, Role-based Access Control and Discretionary access control E) all of the above



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**2024/2025 BSC COMPUTER SCIENCE SECOND SEMESTER EXAMINATIONS
COURSE: COS201 - COMPUTER PROGRAMMING I**

TIME: 1hr

Instruction: Shade the correct option on your OMR Answer sheet

1. _____ is not a feature of Java?
(a) Object-Oriented (b) Platform Dependent
(c) Secure (d) Robust (e) None of the above
2. Java does not support _____.
(a) Multiple inheritance through classes (b) Object-oriented programming (c) Interface implementation (d) Exception handling (e) All of the above.
3. How many primitive data types are there in Java?
(a) 6 (b) 7 (c) 8 (d) 9 (e) 10
4. What will be the output of the following code?

```
int x = 5;
int y = x;
x = 10;
System.out.println(y);
```

 (a) 10 (b) 5 (c) 0 (d) Compilation Error
(e) Syntax Error
5. Which of the following statements about reference types is true? (a) Reference types store values directly (b) Reference types are not objects (c) Reference types store memory addresses (references) (d) Reference types can't be null (e) None of the above
6. Which of the following statements is true regarding procedural vs object-oriented paradigms in Java?
(a) Java does not allow procedural programming because all methods must be non-static.
(b) Procedural programming is discouraged but still syntactically valid in Java.
(c) Procedural programming requires inheritance to implement in Java.
(d) Procedural and object-oriented programming are mutually exclusive in Java (e) None of the above.

7. What procedural programming characteristic is demonstrated by this code?

```
public class Calculator {
    public static int sum(int a, int b) {
        return a + b;
    }
    public static void main(String[] args) {
        int result = sum(10, 20);
        System.out.println(result);
    }
}
```

- (a) Data encapsulation (b) Inheritance and polymorphism (c) Use of static functions without objects (d) Dynamic method dispatch (e) Functional programming.
8. _____ does not align with Java feature of being a **robust language**. (a) Automatic garbage collection (b) Strong memory management (c) Lack of pointer arithmetic (d) Extensive use of goto statements (e) Multithreading
9. Java programs are compiled into an intermediate form called _____. (a) Machine code (b) Source code (c) Bytecode (d) Assembly code (e) Tinycode

Using the code snippet below, answer questions 10 – 11

```
import java.util.Scanner;
public class UserInputExample {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter your name: ");
        String name = scanner.nextLine();
        System.out.println("Hello, " + name);
        scanner.close();
    }
}
```

10. Why is scanner.nextLine() used instead of scanner.next() in this program? (a) nextLine() ignores user input (b) nextLine() reads only the

first word (c) nextLine() reads the entire line including spaces (d) nextLine() is deprecated. (e) Don't Know why.

11. Which package must be imported to use the Scanner class in Java? (a) java.io (b) java.util (c) java.lang (d) java.input (e) java.string

12. What is variable shadowing in Java? (a) When two methods have the same name but different parameters (b) When a subclass defines a field with the same name as in its superclass (c) When a constructor calls another constructor in the same class (d) When a local variable is declared inside a loop (e) None of the above

13. Which cast is allowed without an explicit cast operator? (a) int to long (b) long to int (c) Object to String (d) double to float (e) object to int

14. What is method overloading in Java? (a) Defining multiple methods with the same name and same parameters (b) Defining methods with different names (c) Defining multiple methods with the same name but different parameters (d) Using one method in another class (e) None of the above

Consider the code snippet below and answer questions 15 to 17

```
1 public class Dog {
2     int Height;
3     private int weight;
5     public Dog() {
6         weight = 42;
7     }
9     public void cat(){
10
11     }
13 }
```

15. How many method(s) does the class Dog contain? (a) 1 (b) 2 (c) 3 (d) 4 (e) 5

16. What is the Access Specifier of the variable Height? (a) Public (b) Private (c) Protected (d) None of the Above (e) All of the Above

17. Which of the following is true of the class Dog?
I) The name of the file is DOG.txt
II) The name of the file is unknown
III) The class does not have a default constructor
IV) The variable weight is Encapsulated
(a) I Only (b) I and II (c) III (d) III and IV (e) IV only

18. The code below will not run because of an error. What should be done to resolve the issue?

```
int a = 40;
short b = a;
```

(a) Garbage Collection (b) Casting
(c) Overloading (d) Polymorphism
(e) Instantiation

Consider the code below and answer questions 19 to 21:

```
public class Student{
    String Name;
    private int Age;
    Student John;
    void setAge(int Age){
        this.Age=Age;
    }

    char Sex;
    int getAge (){
        Return Age;
    }
    public void createObject() {
        John = new Student();
    }
}
```

19. What type of variable is the variable, John? (a) Local Variable (b) Automatic Variable (c) Instance variable (d) Static Variable (e) None of the Above

20. Which of the following variables is an instance Variable? (a) Name (b) Age (c) Sex (d) All of the Above (e) None of the Above

21. What is the return type of the setAge method. (a) int (b) char (c) String (d) no return type (e) default

22. Identify the return type of a method that does not return any value. (a) int (b) void (c) double (d) return (d) None of the Above

23. What are the variables declared in a class for the use of all methods of the class called? (a) Local Variables (b) Instance variables (c) Reference variable (d) Class variables (e) None of the Above

24. Identify the prototype of the default constructor in the following code:

```
public class Solution {
}
```

(a) Solution(void) (b) void Solution() (c) public Solution(void) (d) public Solution() (e) None of the Above

25. Which one of the following is not an access specifier? (a) Protected (b) Static (c) Public (d) Private (e) None of the Above

26. Which of the following keywords allows a class to achieving multiple inheritance? (a) extends (b) implements (c) inherits (d) uses (e) None of the Above

27. How can you achieve polymorphism in Java? (a) By method overloading (b) By method overriding (c) By both method overloading and overriding (d) By extending classes (e) None of the Above

28. What happens if a method in a subclass has the same name and parameters as a method in its superclass? (a) The superclass method is executed (b) The subclass method is executed (c) A compile-time error occurs (d) The program fails to run (e) The subclass method is overridden

29. Which of the following is valid for single inheritance?

- (a) class SubClass extends SupperClass {}
- (b) Subclass inherits SupperClass {}
- (c) class SubClass implements SupperClass {}
- (d) SubClass : SupperClass {}
- (e) Subclass extends SupperClass {}

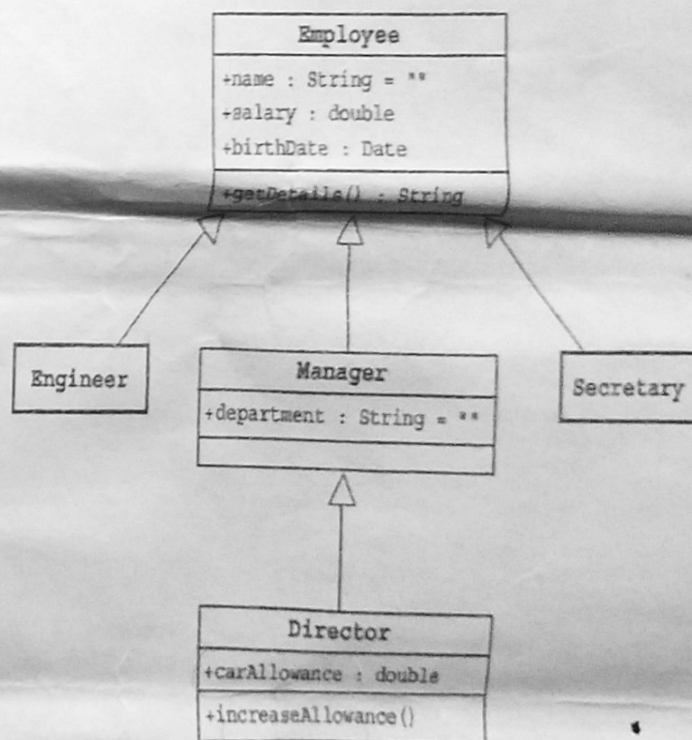
30. Which operator is used to instantiate a class? (a) extends (b) new (c) constructor (d) plus (e) reference

31. Find the output of the following program.

```
public class Solution {
    public static void main(String[] args){
        short x = 10;
        x = x * 5;
        System.out.print(x);
    }
}
```

(a) 50 (b) 10 (c) 15 (d) 0 (e) Compile error

Study the diagram below and answer questions 32 and 33



32. How many variables does the Director class have? (a) 1 (b) 3 (c) 4 (d) 5 (e) 7

33. If the class defines a method, getDetails(), which of the following an overridden method?

- (a) Engineer.getDetails();
- (b) Director.getDetails();
- (c) Employee.getDetails();
- (d) Secretary.getDetails();
- (e) None of the Above

34. In a Java program, you must use _____ to separate classes, objects, and methods. (a) commas (b) semicolons (c) dots (d) forward slashes (e) apostrophe
35. Envisioning program components as objects that are similar to concrete objects in the real world is the hallmark of _____. (a) command-line operating systems (b) procedural programming (c) object-oriented programming (d) machine languages (e) method
36. The values of an object's attributes are known as its _____. (a) state (b) orientation (c) method (d) condition (e) class
37. An instance of a class is an _____. (a) method (b) procedure (c) object (d) case (e) class
38. Languages that let you use an easily understood vocabulary of descriptive terms, such as read, write, or add, are known as _____ languages. (a) procedural (b) machine (c) object-oriented (d) high-level (e) structured
39. The rules of a programming language constitute its _____. (a) logic (b) syntax (c) format (d) objects (e) byte code
40. A _____ translates high-level language statements into machine code. (a) programmer (b) syntax detector (c) compiler (d) decipherer (e) code
41. Named computer memory locations are called _____. (a) compilers (b) addresses (c) appellations (d) variables (e) bus
42. Java is architecturally _____. (a) neutral (b) oriented (c) specifics (d) abstract (e) object
43. All Java programming statements must end with a _____. (a) period (b) comma (c) closing parenthesis (d) semicolon (e) period
44. Arguments to methods always appear within _____. (a) parentheses (b) double quotation marks (c) single quotation marks (d) curly braces (e) square bracket
45. All Java applications must have a method named _____.
(a) method () (b) main () (c) java () (d) Hello () (e) class ()



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2024/2025 BSC COMPUTER SCIENCE FIRST SEMESTER EXAMINATIONS
COURSE: U-INS212 Management Information System
CREDIT UNIT: 03

TIME: 50 minutes

INSTRUCTIONS: Attempt all Questions and shade the option corresponding to the answer chosen.

1. What is the primary objective of a Management Information System (MIS)?
(a) Replace managerial decision-making (b) Support organizational functions and decision-making (c) Increase operational costs (d) Eliminate communication within organizations (e) Store data only for future reference
2. Which activities are involved in transforming raw data into useful information?
(a) Input, processing, output (b) Storage, retrieval, deletion (c) Analysis, filtering, exporting (d) Capturing, encrypting, deleting (e) None of the above
3. Which component of MIS is responsible for collecting raw data from various sources?
(a) Data Storage (b) Data Processing (c) Data Collection (d) Information Output (e) User Interface
4. Which of the following is NOT a type of MIS? (a) Transaction Processing System (TPS) (b) Decision Support System (DSS) (c) Executive Information System (EIS) (d) Human Resource Management System (HRMS) (e) Customer Relationship Management (CRM)
5. What is the role of Decision Support Systems (DSS)? (a) Process daily transactions (b) Assist in making complex decisions using analytical models (c) Provide top-level executives with summaries only (d) Manage customer interactions exclusively (e) Store employee data only
6. Which of the following is an example of a Customer Relationship Management (CRM) system?
(a) SAP ERP (b) Salesforce Chatter (c) Oracle DBMS (d) Google Drive (e) Microsoft Excel
7. Why are complementary assets important for MIS? (a) They replace the need for technology investments (b) They help maximize the value of information systems (c) They reduce the need for data collection (d) They eliminate the need for organizational change (e) They are optional for effective MIS implementation
8. Which of the following is NOT considered a complementary asset? (a) Organizational culture and leadership (b) Business processes (c) Human capital and skills (d) Software licensing fees (e) Infrastructure and technology
9. Which discipline contributes to the technical approach to studying MIS? (a) Sociology (b) Economics (c) Computer Science (d) Psychology (e) Management
10. The sociotechnical systems perspective emphasizes the importance of:
(a) Only technological features (b) Only social features (c) The best fit between social and technical features (d) Ignoring organizational culture (e) Hardware over software
11. Which component of an information system involves converting raw data into meaningful information?
(a) Data Collection (b) Data Storage (c) Data Processing (d) Information Output (e) User Interface
12. Which of the following is an example of a support activity in the value chain?
(a) Inbound logistics (b) Operations (c) Human resources management (d) Outbound logistics (e) Sales and marketing
13. The process of comparing your business processes against industry standards is called:
(a) Benchmarking (b) Outsourcing (c) Standardization (d) Benchmarking (e) Optimization
14. Which of the following best describes a digital firm?
(a) A company that operates solely offline (b) A firm where digital technologies are central to operations and customer interactions (c) A company that does not use internet technologies (d) A traditional brick-and-mortar business with no online presence (e) A firm that only uses manual processes
15. Which of the following is NOT a benefit of being a digital firm?
(a) Enhanced efficiency (b) Improved decision-making (c) Greater customer engagement (d) Increased operational costs (e) Innovation
16. Which element of a business process involves resources needed to perform tasks?
(a) Inputs (b) Activities (c) Outputs (d) Stakeholders (e) Controls
17. Which level of management is responsible for long-term strategic decisions? (a) Senior management (b) Middle management (c) Operational management (d) Data workers (e) Production workers
18. The physical equipment used in an information system is called: (a) Data (b) Software (c) Hardware (d) Network (e) Cloud service
19. Which of the following is a type of network that is accessible only within a company?
(a) Internet (b) Extranet (c) Intranet (d) Web server (e) Cloud network

20. Which of the following is NOT a component of the IT infrastructure? (a) Hardware (b) Software (c) Data management technology (d) Employee training programs (e) Telecommunications technology
21. E-commerce primarily involves: (a) Internal communication within a company (b) Buying and selling of goods/services over the internet (c) Internal management of business processes (d) Government data management (e) Physical retail store transactions only
22. Which platform is known for enabling employees to contribute and edit content easily? (a) Wikis (b) ERP systems (c) CRM systems (d) Data warehouses (e) E-commerce platforms
23. The term "synergies" in a value web context refers to: (a) Independent operations of firms (b) Sharing outputs as inputs to reduce costs and increase profits (c) Reducing communication among partners (d) Eliminating supply chain relationships (e) None of the above
24. Which of the following best explains "core competencies"? (a) Activities performed by entry-level employees (b) Activities in which a firm is a global leader and excels (c) Activities that are outsourced (d) Tasks that are unimportant for competitive advantage (e) All of the above
25. Which strategy involves targeting a specific segment of the market? (a) Low-cost leadership (b) Product differentiation (c) Focus on market niche (d) Customer and supplier intimacy (e) Broad market approach
26. Which of the following is NOT a typical function of MIS? (a) Supporting operational efficiency (b) Facilitating strategic planning (c) Replacing senior management decision-making (d) Performance monitoring (e) Providing decision-making support
27. Which of the following enables organizations to share information and collaborate across locations? (a) E-mail and instant messaging (b) Business Process Modeling Notation (BPMN) (c) Value stream mapping (d) SWOT analysis (e) All of the above
28. Which role is responsible for translating business problems into information requirements? (a) CIO (b) Data entry clerk (c) System analyst (d) End user (e) HR manager
29. The term "extranets" refers to: (a) Internal company Web sites only accessible by employees (b) Public websites open to everyone (c) Company Web sites accessible to external vendors and suppliers (d) The entire internet infrastructure (e) None of the above
30. Which of the following is a benefit of strengthening customer and supplier intimacy? (a) Increased costs (b) Reduced customer loyalty (c) Improved innovation and efficiency (d) Decreased collaboration (e) Less reliable supply chains
31. Which of these is an example of a primary activity in the value chain? (a) Inbound logistics (b) Operations (c) Human resources management (d) Outbound logistics (e) Sales and marketing
32. Which role is responsible for managing the firm's information technology strategy? (a) Chief Security Officer (CSO) (b) Chief Information Officer (CIO) (c) Chief Privacy Officer (CPO) (d) Chief Financial Officer (CFO) (e) Human Resources Manager
33. Which of the following best describes a "value web"? (a) A single company's internal workflow system (b) A network of interconnected firms that coordinate their processes to respond rapidly to supply and demand changes (c) A traditional supply chain (d) A marketing strategy for online advertising (e) None of the above
34. Which system is primarily used to handle daily transactions like billing? (a) Decision Support System (DSS) (b) Transaction Processing System (TPS) (c) Executive Information System (EIS) (d) Customer Relationship Management (CRM) (e) Business Intelligence (BI) System
35. Which of the following is a key benefit of using MIS for strategic planning? (a) Increased operational costs (b) Better insights into trends and forecasts (c) Eliminating decision-making (d) Reducing data collection efforts (e) None of the above
36. Which component of an information system encompasses data management technology? (a) Hardware (b) Software (c) Data management technology (d) Network (e) User interface
37. Which of the following best explains "mass customization"? (a) Producing identical products for all customers (b) Customizing products using the same production resources as mass production to meet individual customer needs (c) Standardizing all products for efficiency (d) Outsourcing all custom work (e) None of the above
38. What is the main role of a systems analyst? (a) Write code for applications (b) Translate business requirements into information systems (c) Manage hardware infrastructure (d) Oversee data entry tasks (e) Handle marketing strategies
39. The dominant global network that links millions of networks is called: (a) LAN (b) Intranet (c) Internet (d) Extranet (e) VPN
40. The practice of comparing your business processes against industry standards is called: (a) Benchmarking (b) Outsourcing (c) Data mining (d) Standardization (e) Process reengineering



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2024/2025 BSC COMPUTER SCIENCE FIRST SEMESTER EXAMINATIONS

Course Code: DTS201 - Introduction to Data Science

CREDIT UNIT: 02

TIME: One Hour

Instruction: Answer ALL Questions

Shade right options with HB Pencil

Section A: Fundamentals of Data Science

1. Which of the following best describes the primary goal of data science?
a) Data storage optimization
b) Developing user interfaces
c) Extracting insights from data
d) Designing hardware components
e) Managing network protocols
2. In data science, the term "feature" refers to:
a) A software bug
b) An individual measurable property of a phenomenon
c) A type of data visualization
d) A data storage unit
e) A machine learning algorithm
3. Which discipline is NOT traditionally integrated into data science?
a) Statistics
b) Computer Science
c) Domain Expertise
d) Astrology
e) Mathematics
4. The process of cleaning and organizing raw data into a usable format is known as:
a) Data visualization
b) Data mining
c) Data preprocessing
d) Data warehousing
e) Data encryption
5. Which of the following is a characteristic of big data?
a) Low volume
b) Static structure
c) High velocity
d) Limited variety
e) Minimal veracity
6. The term "data science" was first coined in which decade?
a) 1960s
b) 1970s
c) 1980s
d) 1990s
e) 2000s
7. Which of the following is NOT a step in the data science process?
a) Data collection
b) Data preprocessing
c) Model deployment
d) Software engineering
e) Model evaluation
8. In the context of data science, "EDA" stands for:
a) Electronic Data Analysis
b) Exploratory Data Analysis
c) Enhanced Data Algorithm
d) External Data Access
e) None of the above
9. Which of the following best describes supervised learning?
a) Learning without labeled data
b) Learning with labeled data
c) Learning through reinforcement

- d) Learning via unsupervised methods
- e) None of the above

10. Which of the following is an example of unstructured data?
a) Excel spreadsheet
b) SQL database
c) Text document
d) CSV file
e) Relational table

Section B: Data Extraction and Big Data

11. Which of the following tools is commonly used for web scraping?
a) NumPy
b) BeautifulSoup
c) Matplotlib
d) TensorFlow
e) PyTorch
12. In big data, the "3 Vs" stand for Volume, Velocity, and:
a) Variety
b) Veracity
c) Validity
d) Value
e) Visualization
13. Which of the following is a distributed computing framework for big data processing?
a) Hadoop
b) Excel
c) SPSS
d) Tableau
e) SAS
14. MapReduce is a programming model associated with:
a) Data visualization
b) Big data processing
c) Statistical analysis
d) Machine learning
e) Database management
15. Which of the following is NOT a characteristic of big data?
a) Volume
b) Velocity
c) Variety
d) Veracity
e) Vagueness
16. The process of identifying patterns in large datasets to predict future trends is known as:
a) Data cleaning
b) Data mining
c) Data entry
d) Data storage
e) Data encryption
17. Which of the following is a NoSQL database?
a) MySQL
b) PostgreSQL
c) MongoDB
d) Oracle
e) SQLite
18. In the context of big data, "HDFS" stands for:
a) High-Definition File System
b) Hadoop Distributed File System
c) Hybrid Data File Storage
d) Hierarchical Data File System
e) None of the above
19. Which of the following is a real-time data processing framework?
a) Hadoop MapReduce
b) Apache Spark
c) Microsoft Excel
d) Tableau
e) SPSS

20. Which of the following is NOT a data extraction technique?

- ✓ a) Web scraping
- b) API integration
- c) Data entry
- d) Data visualization
- e) Database querying

Section C: Data Science Tools and Platforms

21. Which programming language is widely used for statistical analysis and data visualization?

- ✓ a) Java
- b) C++
- c) R
- d) Swift
- e) PHP

22. Which Python library is primarily used for data manipulation and analysis?

- ✓ a) NumPy
- b) Pandas
- c) Matplotlib
- d) Seaborn
- e) Scikit-learn

23. Which of the following tools is used for creating interactive data visualizations?

- ✓ a) Jupyter Notebook
- b) Tableau
- c) Notepad++
- d) Eclipse
- e) Git

24. Scikit-learn is a Python library used for:

- ✓ a) Web development
- b) Machine learning
- c) Data visualization
- d) Database management
- e) Network security

25. Which of the following is a cloud-based platform for data science and machine learning?

- ✓ a) Google Colab
- b) Microsoft Word
- c) Adobe Photoshop
- d) VLC Media Player
- e) Skype

26. Which of the following is NOT a Python library?

- ✓ a) NumPy
- b) Pandas
- c) ggplot2
- d) Matplotlib
- e) Seaborn

27. Which tool is commonly used for version control in data science projects?

- ✓ a) Git
- b) Excel
- c) Tableau
- d) PowerPoint
- e) SPSS

28. Which of the following platforms provides a collaborative environment for data science projects?

- ✓ a) GitHub
- b) Dropbox
- c) OneDrive
- d) Google Drive
- e) iCloud

29. Which of the following is a Python-based web framework?

- a) Django
- b) Flask
- ✓ c) Both a and b
- d) React
- e) Angular

30. Which of the following is used for creating dashboards and business intelligence reports?

- ✓ a) Tableau
- b) Excel
- c) Power BI
- d) All of the above
- e) None of the above

Section D: Data Science Processes and Methodologies

31. The process of dividing data into training and testing sets is known as:

- ✓ a) Data splitting
- b) Data shuffling

- c) Data merging
- d) Data cleaning
- e) Data visualization

32. Which of the following is NOT a step in the CRISP-DM process model?

- ✓ a) Business understanding
- b) Data understanding
- c) Data preparation
- d) Model deployment
- e) Data encryption

33. In data preprocessing, handling missing values can be done by:

- ✓ a) Deleting rows with missing values
- b) Replacing missing values with mean or median
- c) Using algorithms that support missing values
- d) All of the above
- e) None of the above

34. Which of the following is a technique for reducing the dimensionality of data?

- ✓ a) Principal Component Analysis (PCA)
- b) Linear Regression
- c) Decision Trees
- d) K-Means Clustering
- e) Naive Bayes

35. Which of the following is a measure of model performance for classification problems?

- ✓ a) Mean Squared Error
- b) R-squared
- c) Accuracy
- d) Adjusted R-squared
- e) None of the above

36. Which of the following is NOT a data preprocessing technique?

- a) Normalization
- b) Standardization
- c) Data augmentation
- ✓ d) Data encryption
- e) Handling missing values

37. The process of transforming raw data into an understandable format is known as:

- ✓ a) Data cleaning
- b) Data preprocessing
- c) Data transformation
- d) Data integration
- e) Data visualization

38. Which of the following is a technique used to prevent overfitting in machine learning models?

- ✓ a) Cross-validation
- b) Underfitting
- c) Data leakage
- d) Data augmentation
- e) None of the above

39. Which of the following is NOT a type of machine learning?

- a) Supervised learning
- b) Unsupervised learning
- c) Reinforcement learning
- d) Deep learning
- ✓ e) None of the above

40. Which of the following is a key difference between data analytics and data science?

- ✓ a) Data analytics is more focused on future predictions, while data science focuses on historical analysis.
- b) Data science is a broader field that includes data analytics, along with aspects of machine learning, statistics, and computer science.
- c) Data analytics deals with structured data, while data science handles unstructured data.
- d) Data science is primarily used in academic research, while data analytics is used in business.
- e) There is no significant difference between data analytics and data science.

34. The key reason for post-implementation reviews is: A. Budget verification B. Performance benchmarking C. Stakeholder negotiation D. Regulatory auditing E. Data center optimization
35. A poor system test data plan most often leads to A. Project scope creep B. User interface inconsistency C. Functional redundancies D. Latent system faults E. Inadequate documentation
36. Which of the following file conversion challenges is most likely to impact data integrity?
A. Index corruption B. Batch job failure C. Operating system mismatch
D. Memory leak E. Multithreaded conflict
37. What is the main responsibility of the data control staff during system installation?
A. Defining new record structures B. Performing test case analysis C. Monitoring data movement and validity D. Managing file permissions E. Writing software patches
38. During installation, data preparation staff are expected to: A. Program data entry screens
B. Configure network protocols C. Reformat and validate data sets
D. Review system logs E. Conduct user evaluations
39. The greatest risk of direct installation strategy is: A. Delayed stakeholder approval
B. High testing overhead C. Lack of fallback system D. Redundant staffing E. Legal compliance
40. Which approach most effectively mitigates data migration risks during installation? A. Load balancing
B. Parallel systems comparison C. Server caching D. Disk mirroring E. Incremental rollout
41. Which of the following would be considered an intangible benefit in system evaluation?
A. Reduced server maintenance cost B. Decreased processing time C. Improved customer satisfaction D. Fewer full-time employees E. Lower electricity bills
42. System evaluation that focuses on regulatory compliance falls under: A. Operational analysis
B. Cost-benefit assessment C. External audit requirement D. Technical validation
E. Strategic review
43. Which evaluation method best assesses system performance under variable load conditions?
A. Focus group feedback B. Regression testing C. Stress testing D. Cost-to-benefit ratio
E. Risk assessment matrix
44. What is the main purpose of post-implementation recommendations? A. Reduce developer workload
B. Suggest training schedules C. Ensure continuous improvement D. Justify project delays
E. Identify regulatory violations
45. Which system evaluation metric is most susceptible to subjective bias? A. Net present value
B. Usability score C. Throughput rate D. Downtime duration E. Defect density
46. A system that fails to meet an external requirement will likely face: A. Cost escalation B. Increased user training
C. Legal consequences D. Storage bottlenecks E. Redundant scripting
47. Which scenario represents a negative net benefit in cost-benefit analysis? A. Training costs outweigh efficiency gains
B. Revenue increase after system adoption C. Maintenance costs decrease over time D. ROI greater than 1 E. Break-even in under a year
48. Which of the following best supports quantitative evaluation of systems success? A. Employee morale surveys
B. Incident report reviews C. Metrics on processing delays D. Client testimonials
E. Stakeholder interviews
49. A benefit realization plan is used in evaluation to: A. Document project risks B. Translate outputs into measurable outcomes
C. Allocate task resources D. Align team roles with deliverables
E. Approve technical constraints
50. A key limitation of cost-benefit analysis in systems evaluation is: A. High level of automation
B. Lack of documentation C. Inability to capture long-term impacts accurately D. Ease of duplication E. Requirement for software modeling

- new system operation cost C. Vendor vs. In-house development cost D. Licensing vs. subscription cost E. UI design vs. backend optimization cost
17. Feasibility analysis often omits in early-stage decision-making.
A. Legal feasibility B. Emotional resistance from users C. Budgetary constraints
D. Timelines for development E. Technological compatibility
 18. Which feasibility type is best evaluated using Monte Carlo simulation or probabilistic forecasting?
A. Operational B. Schedule C. Technical D. Legal E. Economic
 19. Which DFD level represents the most granular view of system processes?
A. Level-0 B. Level-1 C. Level-2 D. Context diagram E. Entity-Relationship Diagram
 20. Which of the following is a direct consequence of incorrect analysis specifications?
A. Inconsistent documentation B. Accurate test data C. Optimized file conversion
D. Accelerated system evaluation E. Enhanced network throughput
 21. The primary role of a feasibility report is to:
A. Describe coding architecture B. Provide user manuals C. Justify the system investment
D. Design ER diagrams E. Test data accuracy
 22. Which fact-finding technique yields the least biased data?
A. Interviews B. Questionnaires C. Observation D. Prototyping E. Document analysis
 23. Which type of requirement is typically identified after user shadowing? A. Functional
B. Operational C. Technical D. Non-functional E. Regulatory
 24. Which of the following is a major limitation of document review in fact-finding?
A. High cost B. Subjectivity C. Time inefficiency D. Inaccuracy due to outdated information
E. Limited scalability
 25. In requirements engineering, traceability primarily helps in: A. User feedback analysis
B. Defining system boundaries C. Ensuring complete documentation D. Linking needs to deliverables
E. Training stakeholders
 26. What is the most significant challenge in producing technical documentation for system design?
A. Writing in natural language B. Keeping documentation up-to-date C. Visualizing user interfaces
D. Translating pseudocode E. Explaining data types
 27. Which system specification task is most dependent on prior feasibility findings?
A. Interface layout B. Logic flow design C. Process modeling D. Hardware platform choice
E. Access control configuration
 28. Which diagram is least useful for specifying logical process design? A. Flowchart B. Pseudocode
C. State diagram D. ER Diagram E. DFD
 29. Which aspect of documentation is most critical for system maintainers? A. Marketing documentation
B. User interface wireframes C. System architecture blueprints D. Help manuals E. Deployment checklists
 30. What ensures consistency between design documentation and implementation? A. Regular code refactoring
B. Version control of diagrams C. Continuous integration D. Traceability matrix
E. Data dictionary
 31. In system implementation, which factor is most crucial when planning parallel runs? A. Hardware availability
B. User resistance level C. Data redundancy D. System load forecasting E. Cost of dual operation
 32. What is the most strategic function of the Education Coordinating Committee during system implementation? A. Approving budget allocations
B. Configuring hardware modules C. Managing user training consistency D. Writing user manuals
E. Designing the UI
 33. Which of the following best exemplifies a white-box test case during system test data preparation?
A. Validating UI flow for errors B. Stress-testing a login form C. Inputting invalid data to check crash behavior
D. Verifying loop conditions and branches E. Analyzing black swan scenarios



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2024/2025 BSC COMPUTER SCIENCE FIRST SEMESTER EXAMINATIONS

COURSE: INS204 – SYSTEM ANALYSIS AND DESIGN

CREDIT UNIT: 03

TIME: 1 hour

INSTRUCTION: Answer all questions

1. Which of the following best defines a system? A. A random collection of components
B. A group of independent entities C. A set of interrelated components working towards a goal
D. A file structure used for programming E. A user interface
2. Which of these is NOT a common type of system? A. Manual B. Automated C. Hybrid
D. Transparent E. Open
3. Which phase of the SDLC involves determining the needs of the end user? A. Design
B. Implementation C. Requirements analysis D. Testing E. Deployment
4. A major objective of system approach is to: A. Eliminate all human roles
B. Focus only on technical requirements C. View problems holistically
D. Reduce the number of components E. Avoid feedback
5. A non-functional requirement includes: A. User must log in with a password
B. System must support login authentication C. System should respond within 2 seconds
D. Sales data must be stored E. Database should contain 10 tables
6. Which of the following is NOT part of system requirements? A. Functional requirements B. Legal constraints
C. Physical location of the user D. Hardware interface E. Organizational structure
7. A level-0 DFD is also known as: A. Primitive DFD B. Abstract DFD C. Context diagram D. Flowchart
E. Control diagram
8. External evaluation might involve: A. Internal managers B. Programmers C. System users only
D. Independent auditors E. Database designers
9. Which of the following is NOT a benefit typically evaluated? A. Time saving B. User satisfaction
C. Developer interest D. Cost reduction E. Flexibility
10. One of the first steps in system implementation is: A. Launching the marketing campaign B. Creating test data
C. Developing a plan D. Interviewing users again E. Deleting old files
11. A deterministic system is defined by: A. Multiple simultaneous outputs for a single input
B. High dependency on environmental feedback C. Predictable behavior under specified inputs
D. Adaptive real-time learning modules E. Self-regulating feedback loops
12. Which SDLC model is best suited for systems with rapidly changing requirements? A. Waterfall
B. Spiral C. V-model D. Big Bang E. Agile
13. A technical requirement can be differentiated from a system requirement by its: A. Broad organizational focus
B. Abstract goal orientation C. Precise implementation details D. Subjectivity to stakeholder interpretation
E. Legal and regulatory conformance
14. Which phase of SDLC most critically depends on a complete understanding of non-functional requirements? A. Design B. Maintenance C. Analysis D. Testing E. Deployment
15. A feasibility study that assesses whether the organization has the capacity to support a new system is: A. Technical B. Operational C. Economic D. Legal E. Schedule
16. Which of the following cost comparisons best helps in determining Return on Investment (ROI) for a proposed system? A. Development vs. maintenance cost B. Current system operation cost vs.